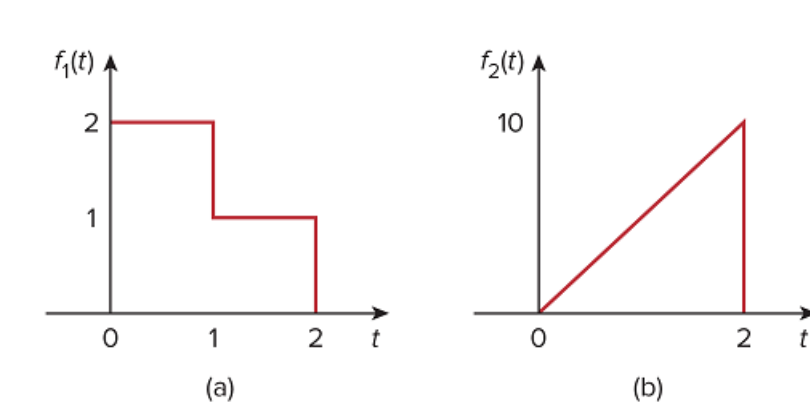


Problem

Find the Fourier transforms of the signals in Fig. 18.32.

Figure 18.32



Step-by-step solution

Step 1 of 6

(a)

Refer to Figure 18.32 (a) in the textbook.

Consider the signal $f_{1a}(t)$ as shown in Figure 1.

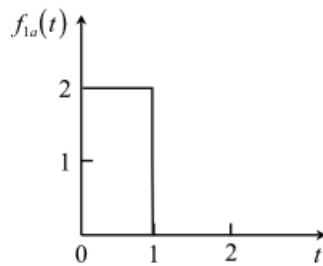


Figure 1

Step 2 of 6

Write the mathematical expression of the signal shown in Figure 1 in terms of step function.

$$f_{1a}(t) = 2[u(t) - u(t-1)] \dots\dots (1)$$

Consider the signal shown in Figure 2.

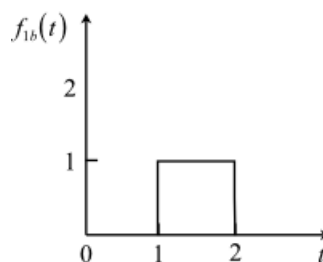


Figure 2

Step 3 of 6

Write the mathematical expression of the signal shown in Figure 2 in terms of step function.

$$f_{1b}(t) = u(t-1) - u(t-2) \dots\dots (2)$$

Add equation 1 and equation 2.

$$\begin{aligned} f_1(t) &= f_{1a}(t) + f_{1b}(t) \\ &= 2[u(t) - u(t-1)] + [u(t-1) - u(t-2)] \\ &= 2u(t) - 2u(t-1) + u(t-1) - u(t-2) \\ &= 2u(t) - u(t-1) - u(t-2) \end{aligned}$$

Step 4 of 6

Apply differentiation on both sides.

$$\begin{aligned} f_1'(t) &= \frac{d}{dt} f_1(t) \\ &= \frac{d}{dt} [2u(t) - u(t-1) - u(t-2)] \\ &= 2\delta(t) - \delta(t-1) - \delta(t-2) \end{aligned}$$

Consider the time differentiation property of the Fourier Transform.

$$\begin{aligned} F[f_1'(t)] &= j\omega F_1(\omega) \\ j\omega F_1(\omega) &= F[2\delta(t) - \delta(t-1) - \delta(t-2)] \\ j\omega F_1(\omega) &= 2 - e^{-j\omega} - e^{-j2\omega} \\ F_1(\omega) &= \frac{2 - e^{-j\omega} - e^{-j2\omega}}{j\omega} \end{aligned}$$

Therefore, the Fourier Transform of the signal $f_1(t)$ is $\boxed{\frac{2 - e^{-j\omega} - e^{-j2\omega}}{j\omega}}$.

Step 5 of 6

(b)

Refer to Figure 18.32 (b) in the textbook.

Write the mathematical expression of the signal $f_2(t)$ in terms of step function.

$$f_2(t) = 5t[u(t) - u(t-2)]$$

Determine the Fourier Transform of $f_2(t)$.

$$\begin{aligned} F_2(\omega) &= \int_{-\infty}^{\infty} f_2(t) e^{-j\omega t} dt \\ &= \int_0^2 5te^{-j\omega t} dt \\ &= 5 \left(\left[t \int e^{-j\omega t} dt \right]_0^2 - \int_0^2 \left(\frac{d(t)}{dt} \right) \frac{e^{-j\omega t}}{(-j\omega)} dt \right) \\ &= 5 \left\{ \left(\frac{te^{-j\omega t}}{-j\omega} \right)_0^2 - \left[\frac{e^{-j\omega t}}{(-j\omega)^2} \right]_0^2 \right\} \end{aligned}$$

Step 6 of 6

Simplify further.

$$\begin{aligned}
 F_2(\omega) &= 5 \left(\frac{te^{-j\omega t}}{-j\omega} \right)_0^2 + 5 \left(\frac{e^{-j\omega t}}{\omega^2} \right)_0^2 \\
 &= 5 \left(\frac{te^{-j\omega t}}{-j\omega} \right)_0^2 + 5 \left(\frac{e^{-j\omega t}}{\omega^2} \right)_0^2 \\
 &= 5 \left[\left(\frac{2e^{-j2\omega}}{-j\omega} \right) - \left(\frac{0e^{-j0\omega}}{-j\omega} \right) \right] + 5 \left[\left(\frac{e^{-j2\omega}}{\omega^2} \right) - \left(\frac{e^{-j0\omega}}{\omega^2} \right) \right] \\
 &= \frac{10e^{-j2\omega}}{-j\omega} + \frac{5}{\omega^2} (e^{-j2\omega} - 1)
 \end{aligned}$$

$$\begin{aligned}
 F_2(\omega) &= \frac{10e^{-j2\omega}}{-j\omega} + \frac{5}{\omega^2} e^{-j2\omega} - \frac{5}{\omega^2} \\
 &= \frac{5}{\omega^2} e^{-j2\omega} + j \frac{10e^{-j2\omega}}{\omega} - \frac{5}{\omega^2} \\
 &= \frac{5}{\omega^2} e^{-j2\omega} (1 + j\omega 2) - \frac{5}{\omega^2}
 \end{aligned}$$

Therefore, the Fourier Transform $f_2(t)$ is $\boxed{\frac{5}{\omega^2} e^{-j2\omega} (1 + j\omega 2) - \frac{5}{\omega^2}}$.